A PROJECT TITLE

ON

Loan Default Prediction Using Random Forest and Gradient Boosting

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF Computer Science and Enegeneering

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1. Abstract:

Loan default is a major challenge for banks and financial institutions because inaccurate assessment of a borrower's repayment ability can result in financial losses. Traditional loan assessment methods often depend on fixed rules and manual analysis, which may not effectively identify complex patterns in borrower and loan information. This project aims to develop a machine learning-based system for predicting whether a borrower is likely to default on a loan. The project will implement and compare two ensemble machine learning algorithms, Random Forest and Gradient Boosting, to determine their effectiveness in loan default prediction. The Lending Club Loan Dataset will be used for this project. It contains relevant borrower and loan information, including loan amount, interest rate, annual income, employment details, credit history, and loan status. The dataset will be preprocessed by handling missing values, encoding categorical attributes, selecting relevant features, and dividing the data into training and testing sets. Random Forest will be used because it can handle complex relationships among features and reduce overfitting by combining multiple decision trees. Gradient Boosting will be used to improve prediction performance by sequentially learning from errors made by previous models. The performance of both models will be evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Among these metrics, recall and ROC-AUC will receive particular attention because correctly identifying potential loan defaulters is important for effective financial risk management. The final results will provide a comparison of Random Forest and Gradient Boosting and identify the better-performing model for predicting loan defaults. The proposed system can help financial institutions make more informed lending decisions, improve credit risk assessment, and reduce potential losses caused by loan defaults.

2. INTRODUCTION:

The increasing number of loan applications has made it difficult for financial institutions to accurately assess the creditworthiness of every borrower using traditional methods. Loan default can cause significant financial losses and can also affect the overall stability and risk management of lending institutions. Therefore, there is a strong motivation to develop an intelligent system that can identify borrowers who are more likely to default before financial decisions are made. Machine learning provides an effective approach for analyzing large amounts of historical loan data and discovering patterns that may not be easily identified through manual analysis. By using machine learning techniques, financial institutions can improve their risk assessment process, make better lending decisions, and reduce the possibility of financial losses.

Loan default prediction is a classification problem in which the objective is to determine whether a borrower is likely to repay a loan or become a defaulter. This project focuses on using Random Forest and Gradient Boosting algorithms to predict loan defaults. The Lending Club Loan Dataset will be used, containing information such as loan amount, interest rate, annual income, employment details, credit history, and loan status. Before model development, the dataset will be cleaned and preprocessed by handling missing values, encoding categorical variables, and selecting relevant features. Random Forest uses multiple decision trees to produce a more reliable prediction, while Gradient Boosting builds models sequentially to improve prediction performance by reducing previous errors. Both algorithms will be trained and tested using the prepared dataset, and their performance will be compared using accuracy, precision, recall, F1-score, and ROC-AUC. The comparison will help identify which algorithm provides better performance for loan default prediction and can be more useful for financial risk assessment.

3. SOFTWARE REQUIREMENTS:

The proposed project requires a computer system with sufficient processing power and memory to perform data preprocessing, machine learning model training, testing, and evaluation. Python will be used as the primary programming language because it provides suitable libraries for data analysis, visualization, and machine learning. Jupyter Notebook or Google Colab can be used as the development environment. Libraries such as Pandas and NumPy will be used for data manipulation and preprocessing, while Scikit-learn will be used to implement the Random Forest and Gradient Boosting algorithms and calculate evaluation metrics. Matplotlib and Seaborn can be used to visualize the dataset and compare model performance. The Lending Club Loan Dataset will be used as the source of data for developing and evaluating the proposed system. A stable internet connection is recommended for downloading the dataset, installing required libraries, and using cloud-based platforms such as Google Colab.

Category	**Requirement**
Operating System	Windows 10/11, Linux, or macOS
Processor	Intel Core i3 or equivalent and above
RAM	Minimum 4 GB; 8 GB recommended
Storage	Minimum 10 GB free space
Programming Language	Python 3.x
Development Environment	Jupyter Notebook / Google Colab
Data Processing Libraries	Pandas, NumPy
Machine Learning Library	Scikit-learn
Visualization Libraries	Matplotlib, Seaborn
Dataset	Lending Club Loan Dataset
Internet Connection	Required for dataset access and online tools

Signature of the Guide

Course Instructor